

Road Network Extraction

from High-Resolution Satellite Imagery ↗



Group 27

Problem Statement ↗



Rapid urbanization demands up-to-date geospatial data, yet manual mapping of transportation infrastructure is labor-intensive. This project addresses the challenge of automated road network extraction from high-resolution satellite imagery. By implementing advanced image processing and feature extraction techniques, the objective is to accurately differentiate road pixels from complex backgrounds and convert them into structured, linear vector layers for seamless GIS integration and urban planning.





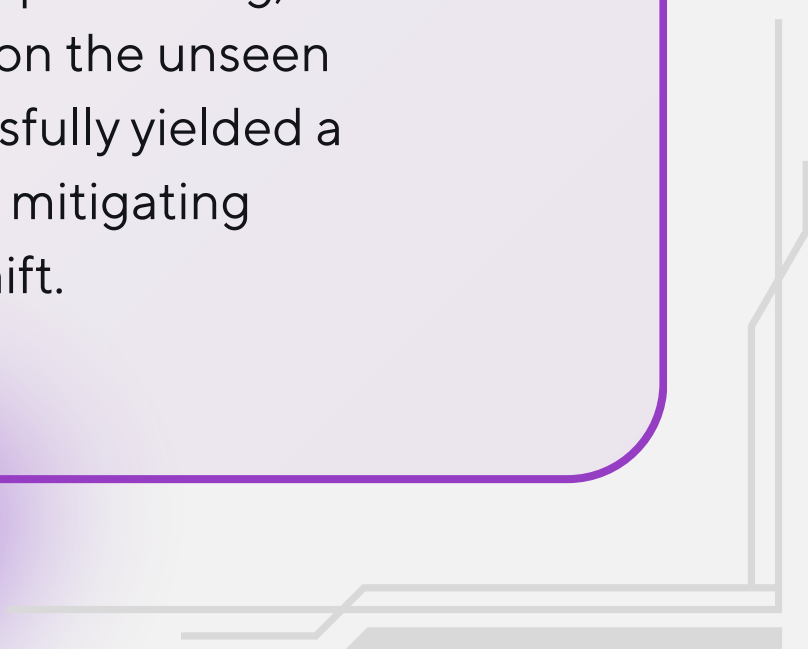
Project Outline ↗

Phase 1: Global Baseline Training

A U-Net architecture with a ResNet-50 encoder was trained on the DeepGlobe benchmark. Utilizing data augmentation and a composite BCE-Dice loss, this 60-epoch stage established a robust general road extraction baseline.

Phase 2: Domain Adaptation & Validation

The baseline was fine-tuned on Indian urban imagery. Applying Test-Time Augmentation and morphological post-processing, zero-shot evaluation on the unseen Indore dataset successfully yielded a 65.69% IoU score, mitigating domain shift.



Dataset Strategy ↗



Stage 1: Global Baseline

- Dataset: DeepGlobe Benchmark (~10,500 patches).
- Purpose: Establish a general understanding of road topology.

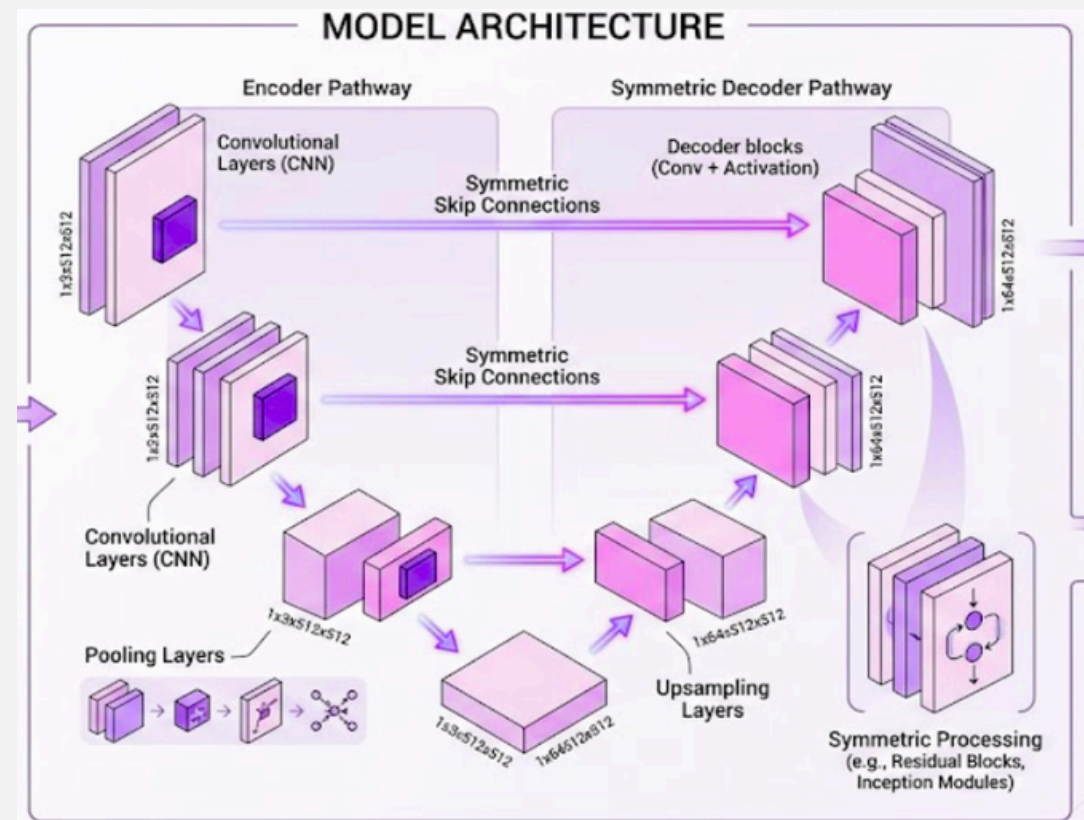
Stage 2: Domain Adaptation

- Dataset: High-resolution imagery from 5 Indian cities (Bhopal, Jabalpur, Gwalior, Ujjain, Nagpur).
- Purpose: Regional fine-tuning to mitigate domain shift.

Stage 3: Zero-Shot Evaluation

- Dataset: Held-out imagery from Indore.
- Purpose: Pure, unbiased benchmark of the final model.

Model Architecture ↗



U-Net with ResNet-50 Encoder

- **Encoder (ResNet-50):** Pre-trained on ImageNet-1K, providing strong initialisation of general visual features and reducing the volume of task-specific data required.
- **Decoder with Skip Connections:** Sequentially up-samples feature maps back to input resolution. Skip connections at each hierarchical level concatenate high-resolution encoder features into the decoder, preserving fine geometric boundaries critical for thin road delineation.

We also tried different model architectures like SegFormer-B3 but it resulted in around same evaluation metrics.

Loss Function ↗

Total Loss: $0.5 \times \text{BCE} + 0.5 \times \text{Soft Dice}$

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\sigma(x_i)) + (1 - y_i) \log(1 - \sigma(x_i))]$$

Where:

N: Total number of pixels in the batch.

y_i : The ground truth binary label (1 for road, 0 for background).

$\sigma(x_i)$: The predicted probability of the pixel being a road.

$$\mathcal{L}_{Dice} = 1 - \frac{2 \cdot TP + 1}{2 \cdot TP + FP + FN + 1}$$

Given $p = \sigma(x)$:

- $TP = \sum (p \cdot y)$
- $FP = \sum (p \cdot (1 - y))$
- $FN = \sum ((1 - p) \cdot y)$

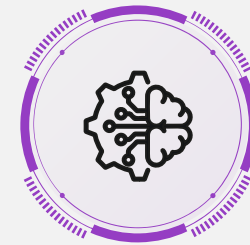
The +1 in the numerator and denominator acts as a smoothing factor to prevent division-by-zero errors on empty patches, ensuring gradient stability.

Training Parameters (baseline/domain) ↗



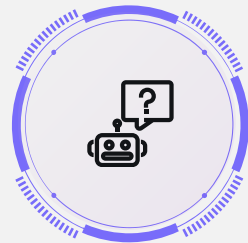
Patch Size (512 x 512)

Captures sufficient spatial context while maintaining manageable computational memory requirements.



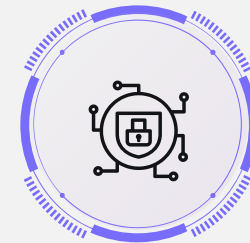
Optimizer (AdamW)

Decouples weight decay from gradient updates for better generalization performance.



Learning Rate (1e-4)

Provides a stable initial step size to prevent divergence early.



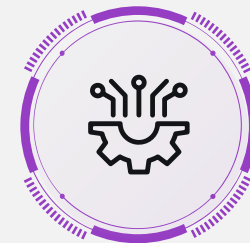
Weight Decay (1e-3)

Regularizes the model by penalizing large weights to reduce overfitting.



Batch Size (8)

Balances memory constraints with stable gradient estimation during feature extraction.



Epochs (60/30)

Allows sufficient iterations to establish a generalized global road baseline.

Image Processing & Post-Processing

Morphological Post-Processing

1. **Morphological Closing**: A dilation followed by erosion bridges fragmented road segments.
2. **Small-Component Removal**: Isolated noise blobs below a minimum pixel threshold are removed

01

Geometric Transforms ↗

Orthogonal rotations and elastic deformations to simulate varied road orientations and terrain distortions.

02

Photometric Adjustments ↗

Hue-saturation shifts to improve robustness against seasonal and lighting variability in satellite imagery.

03

Test-Time Augmentation (TTA) ↗

Multiple spatial permutations (rotations & flips) evaluated at inference; probability maps averaged for stabilized predictions.

Evaluation ↗

DeepGlobe

Evaluated on its native DeepGlobe test split, the initial baseline model demonstrated strong general road extraction capabilities, achieving an IoU of 66.66% and an F1 Score of 0.7847

IoU: 66.66%
F1: 0.78

Indore (baseline training)

When the DeepGlobe-trained model was applied directly to the unseen Indian topology (Indore) before any regional training, performance degraded significantly

IoU: 42.12%
F1: 0.59

Indore (domain adaption)

Following targeted fine-tuning on regional data, the final model successfully overcame the domain gap. Evaluated on the held-out Indore test set, it recovered its performance.

IoU: 65.69%
F1: 0.79

Indore vector road mask



Indore map

Predicted road map

Ground Truth

[result_drive_link](#)

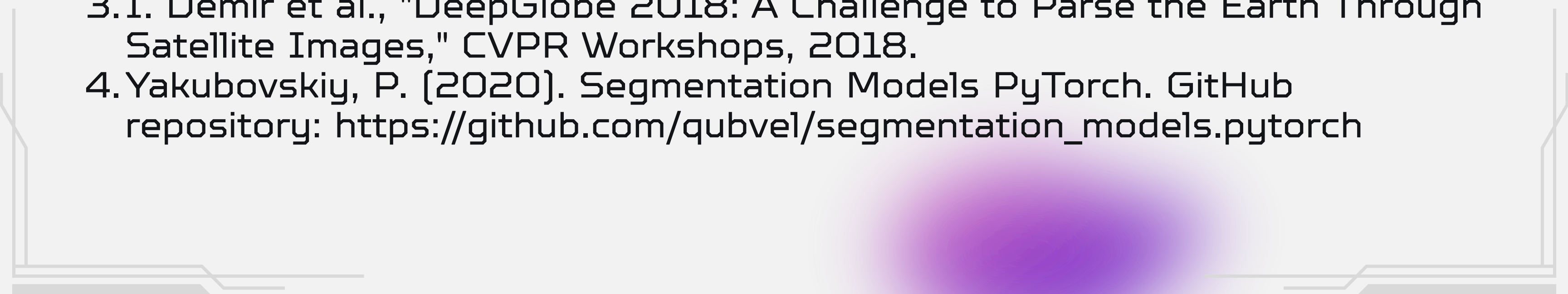
Conclusion & Team Contributions ↗

- **Successful Pipeline Development:** The project successfully established a complete road extraction pipeline, effectively bridging the gap between global baseline training (DeepGlobe) and localized, real-world application (Indian urban datasets).
- **Validation of Domain Adaptation:** The quantitative results strongly validate the necessity of domain adaptation. Regional fine-tuning was critical, recovering the IoU score from a degraded 42.12% to an impressive 65.69% on a completely unseen, held-out city.

Vikash Kumar Singh	Ayush Kumar
Dataset Preparation & Baseline Training	Fine-Tuning & Inference Pipeline.



References ↗

1. Y. Zhou and T. Tan, "C-UNet: Complement UNet for Remote Sensing Road Extraction with Multi-Scale Dilated Convolutions," IEEE Access, 2024.
 2. A. Paszke, S. Gross, S. Chintala, G. Chanan, E. Yang, Z. DeVito, Z. Lin, A. Desmaison, L. Antiga, and A. Lerer, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," Advances in Neural Information Processing Systems (NeurIPS), 2019.
 3. I. Demir et al., "DeepGlobe 2018: A Challenge to Parse the Earth Through Satellite Images," CVPR Workshops, 2018.
 4. Yakubovskiy, P. (2020). Segmentation Models PyTorch. GitHub repository: https://github.com/qubvel/segmentation_models.pytorch
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Thank
You! ↗

